CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS

Evaluation Date: Fri, Oct 10, 2026, 09:00 AM

Patient ID: CV-2024-8841

Architecture: 57-Feature Four-Model Architecture

PATIENT NAME	PATIENT ID	AGE / BIOLOGICAL SEX	HEIGHT / WEIGHT / BMI
Jane Doe	CV-2024-8841	58 years / Female	165 cm / 68 kg (25.0 kg/m^2)

1. MODEL-ESTIMATED CORONARY OUTCOME PROBABILITIES

OVERALL CAD	LAD ARTERY	LCX ARTERY	RCA ARTERY
78.4%	62.1%	41.3%	35.0%
High Risk	Elevated	Moderate	Low Risk

Model Estimation Notice: These are model-estimated probabilities, not clinically calibrated risk estimates or measured coronary artery narrowing percentages.

2. CLINICAL INPUT MEASUREMENTS SUMMARY

Selected patient measurements from current patient session. Machine learning prediction models evaluate the full standardized 57-feature input vector.

PARAMETER	PATIENT VALUE	GENERAL ADULT REFERENCE (INFORMATIONAL)	INFORMATIONAL STATUS
Blood Pressure (BP, Systolic)	130 mmHg	< 120 mmHg (resting systolic)	${parseFloat(bp) >= 130 ? 'Above standard systolic reference (≥ 130 mmHg, Informational)' : parseFloat(bp) >= 120 ? 'Elevated systolic reference (120–129 mmHg, Informational)' : 'Within standard systolic reference'}
Resting Pulse Rate (PR)	72 bpm	60 – 100 bpm	${parseFloat(pr) >= 60 && parseFloat(pr) <= 100 ? 'Within standard resting range' : 'Outside standard resting range (Informational)'}
Left Ventricular EF (EF-TTE)	55%	≥ 50%	${parseFloat(ef) >= 50 ? 'LVEF ≥ 50% — interpret in clinical context' : 'LVEF < 50% — below reference, interpret in clinical context'}
Fasting Blood Sugar (FBS)	105 mg/dL	70 – 99 mg/dL	${parseFloat(fbs) >= 126 ? 'Above standard fasting threshold (≥ 126 mg/dL, Informational)' : parseFloat(fbs) >= 100 ? 'Above standard fasting range (100–125 mg/dL, Informational)' : 'Within standard fasting range'}
Serum Creatinine (CR)	1.0 mg/dL	0.7 – 1.3 mg/dL	Within displayed adult reference range (Informational)
LDL Cholesterol	112 mg/dL	< 100 mg/dL	${parseFloat(ldl) >= 100 ? 'Above standard reference (≥ 100 mg/dL, Informational)' : 'Within standard reference'}
HDL Cholesterol	48 mg/dL	> 40 mg/dL (M) / > 50 mg/dL (F)	${parseFloat(hdl) < 40 ? 'Below standard reference (Informational)' : 'Within standard reference'}

PARAMETER	PATIENT VALUE	GENERAL ADULT REFERENCE (INFORMATIONAL)	INFORMATIONAL STATUS
Triglycerides (TG)	135 mg/dL	< 150 mg/dL	${parseFloat(tg) >= 150 ? 'Above standard reference (Informational)' : 'Within standard reference'}
Evaluated Model Features	**57 Model Inputs**	Standardized 7-domain feature vector	34 numeric and 23 one-hot encoded inputs

3. MODEL VALIDATION PERFORMANCE & RELIABILITY

Retrospective model evaluation metrics recorded during development using held-out test data. Not independently verified prospective clinical trial performance.

MODEL	TARGET VASCULAR OUTCOME	ACCURACY	F1 SCORE	ROC-AUC
CAD Model (`cad_model.pkl`)	Overall Coronary Artery Disease	**85.2%**	89.7%	**0.917**
LAD Model (`lad_model.pkl`)	Left Anterior Descending Stenosis	**65.6%**	67.7%	**0.756**
LCX Model (`lcx_model.pkl`)	Left Circumflex Stenosis	**57.4%**	53.6%	**0.584**
RCA Model (`rca_model.pkl`)	Right Coronary Artery Stenosis	**60.7%**	50.0%	**0.652**

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Model Reliability Notice: LCX and RCA models have limited predictive discrimination in current validation results (ROC-AUC 0.584 and 0.652). Predictions should be interpreted cautiously and must not guide clinical care independently.

4. EDUCATIONAL GUIDANCE & CLINICAL CONSIDERATIONS

Scope of Evaluation & Clinical Considerations:

- Predictions represent statistical model estimates derived from patient input vectors and are not clinical diagnoses.
- Results require interpretation by a qualified physician in full clinical context.
- Actual coronary anatomy and luminal patency cannot be confirmed from machine learning probabilities alone.
- Diagnostic testing and management decisions depend on patient symptoms, physical examination, comprehensive history, and professional medical assessment.
- This application is a research prototype and is not intended for independent clinical decision-making or therapy initiation.

RESEARCH PROTOTYPE — NOT A MEDICAL DEVICE:
For educational and research demonstration purposes only. Predictions must not independently guide clinical decisions. This application is not certified for diagnostic use or treatment planning. Model probabilities and classifications do not substitute for formal diagnostic coronary imaging, physician examination, or qualified medical judgment.